

CS-334: Machine Learning

Lecture 03: Linear Classification

(with In-Class Exercise #3)

09/03/2026

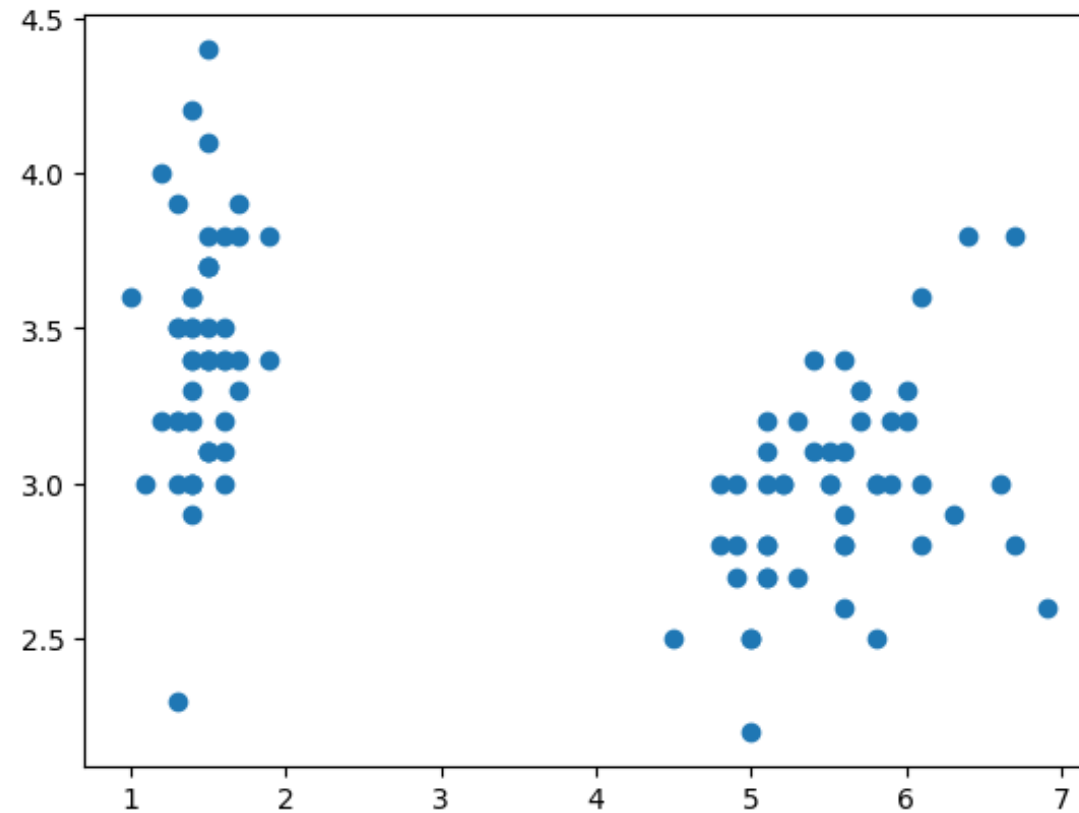
Prof. Shengpu Tang



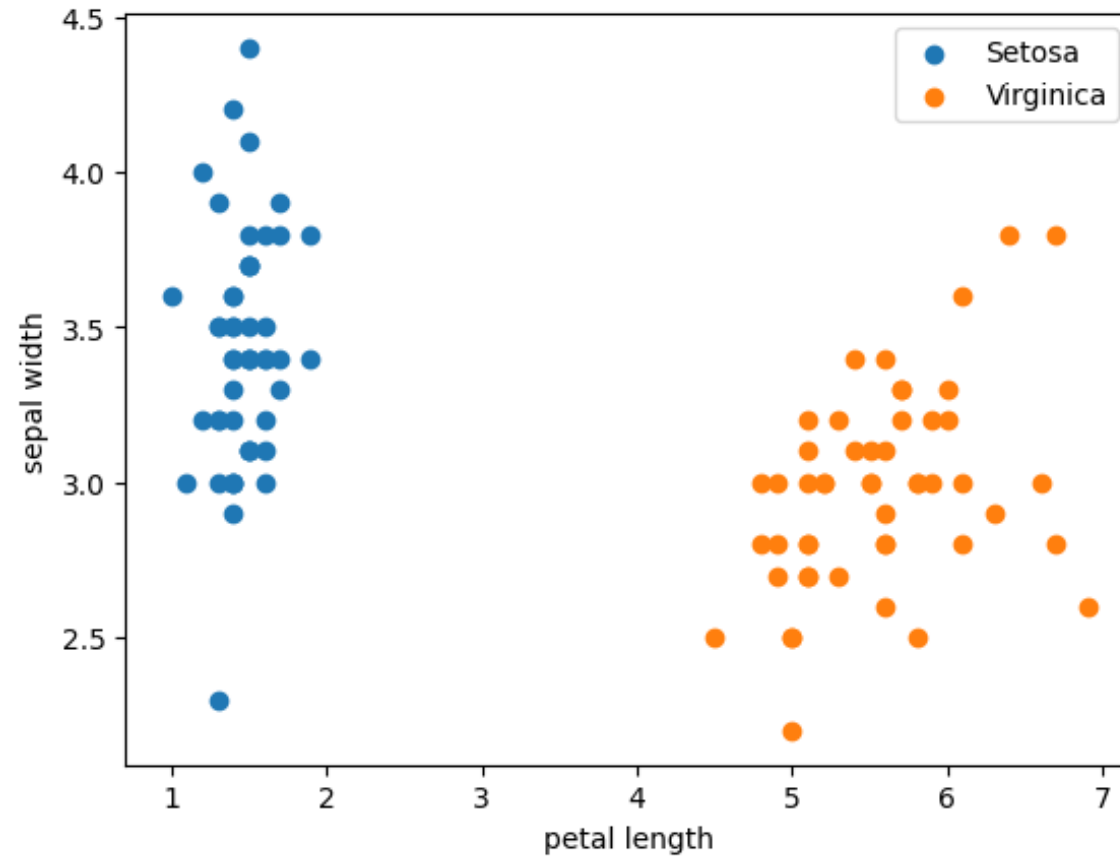
Announcements

- **Linear Algebra + Multivariable Calculus** notes – on Canvas
- **HW1** out; due next Sun 11:59pm on Gradescope
 - Autograder: you can submit multiple times
- **Regular Office Hours** – see canvas for updates
 - My OH: Tue/Thu after class 2:15-3:15pm, or by appt

Recap



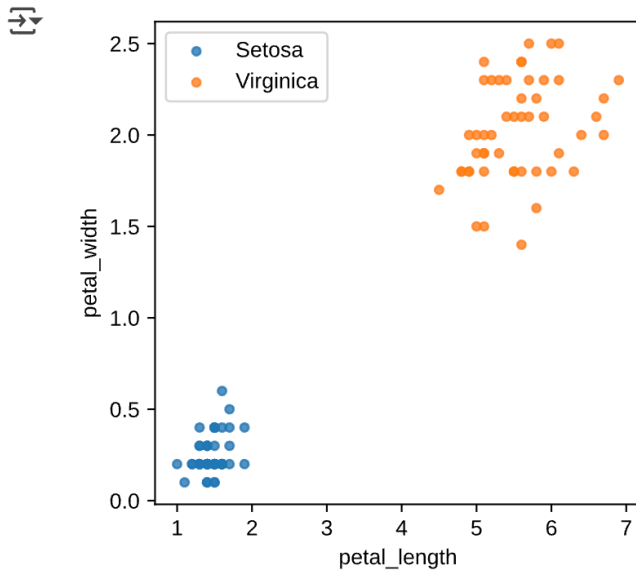
Recap



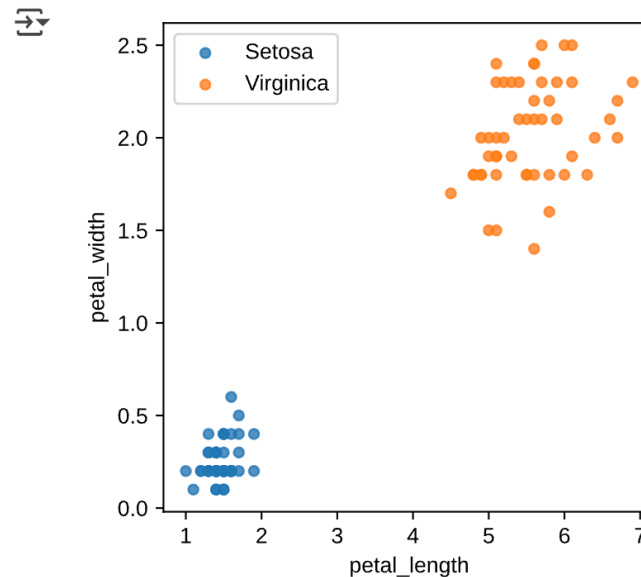
Sample Solution Code - on Canvas

```
# Q4.2 Scatterplot
fig, ax = plt.subplots(figsize=(4,4))
for variety in df['variety'].unique():
    df_variety = df[df['variety'] == variety]
    plt.scatter(df_variety['petal_length'],
                df_variety['petal_width'],
                s=15, alpha=.8, label=variety)

plt.legend()
plt.xlabel('petal_length')
plt.ylabel('petal_width')
plt.show()
```

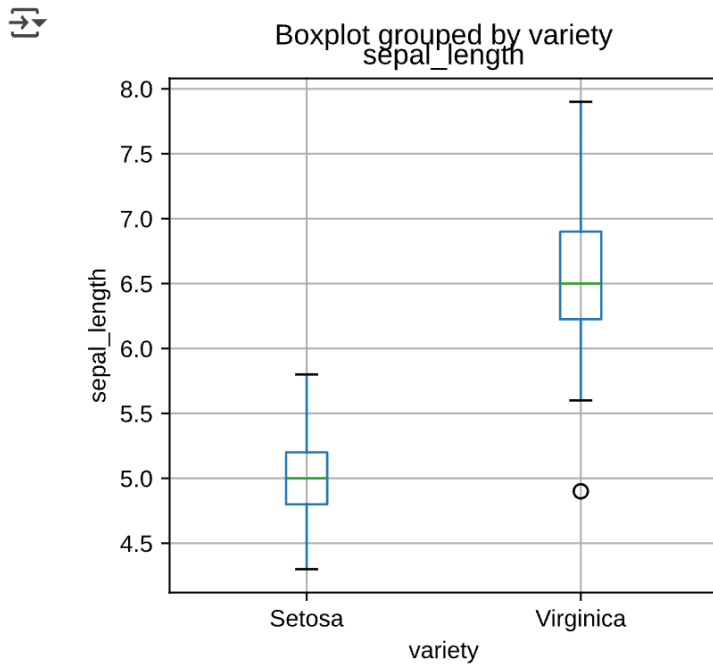


```
# Q4.2 Scatterplot
fig, ax = plt.subplots(figsize=(4,4))
df[df['variety'] == 'Setosa'].plot.scatter(
    x='petal_length', y='petal_width', c='tab:blue', alpha=.8, label='Setosa', ax=ax)
df[df['variety'] == 'Virginica'].plot.scatter(
    x='petal_length', y='petal_width', c='tab:orange', alpha=.8, label='Virginica', ax=ax)
plt.legend()
plt.xlabel('petal_length')
plt.ylabel('petal_width')
plt.show()
```



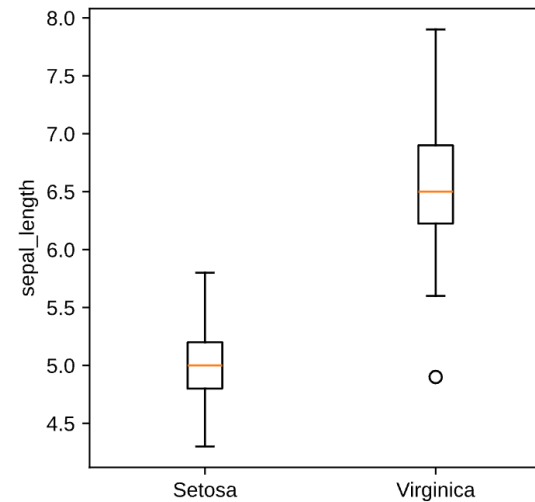
Sample Solution Code - on Canvas

```
# Q4.3 Boxplot
fig, ax = plt.subplots(figsize=(4,4))
df.boxplot(column='sepal_length', by='variety', ax=ax)
plt.ylabel('sepal_length')
plt.show()
```



```
# Q4.3 Boxplot
fig, ax = plt.subplots(figsize=(4,4))
varieties = sorted(df['variety'].unique())
plt.boxplot(
    [df[df['variety'] == variety]['sepal_length'] for variety in varieties],
    labels=varieties,
)
plt.ylabel('sepal_length')
plt.show()
```

<ipython-input-9-f8ac28197762>:4: MatplotlibDeprecationWarning: The 'labels' parameter of plt.boxplot() is deprecated.



Lecture 3: Linear Classification

- feature vector $\vec{x} = [x_1, x_2, \dots, x_d]^T$ $\vec{x} \in \mathbb{R}^d$
- label $y \in \{-1, +1\}$ binary
- training set of labeled examples
 $D = \{ \vec{x}^{(i)}, y^{(i)} \}_{i=1}^N$
- classifier $h: \mathbb{R}^d \rightarrow \{-1, +1\}$

Goal: select the best h from a set of possible classifiers $\mathcal{H}$
that would have the best chance of classifying new examples
 $\rightarrow$ solved by a learning algorithm, typically an optimization problem wrt D .

Example: ECG data sampled at 360 Hz.



let $\vec{x} = [x_1, x_2, \dots, x_{360}]^T$, $x_k \in \{1, \dots, 2048\}$

Suppose training set $N=100$, $D = \{ \vec{x}^{(i)}, y^{(i)} \}_{i=1}^{100}$
in which x_6 is different in every example.

Trivial solution: make a lookup table.

$$h(\vec{x}) = \begin{cases} y^{(i)} & \text{if } x_6 = x_6^{(i)} \\ ? & \text{otherwise} \end{cases}$$

Is this a good classifier? no, it overfits

Generalization $\rightarrow$ works well on unseen examples

- Problem: too many choices in $\mathcal{H}$
- Solution(?): constrain $\mathcal{H}$. but... can't be too "small" or we'll end up not even able to classify D either.

Finding the right balance $\rightarrow$ model selection

Linear Classifier (through the origin)

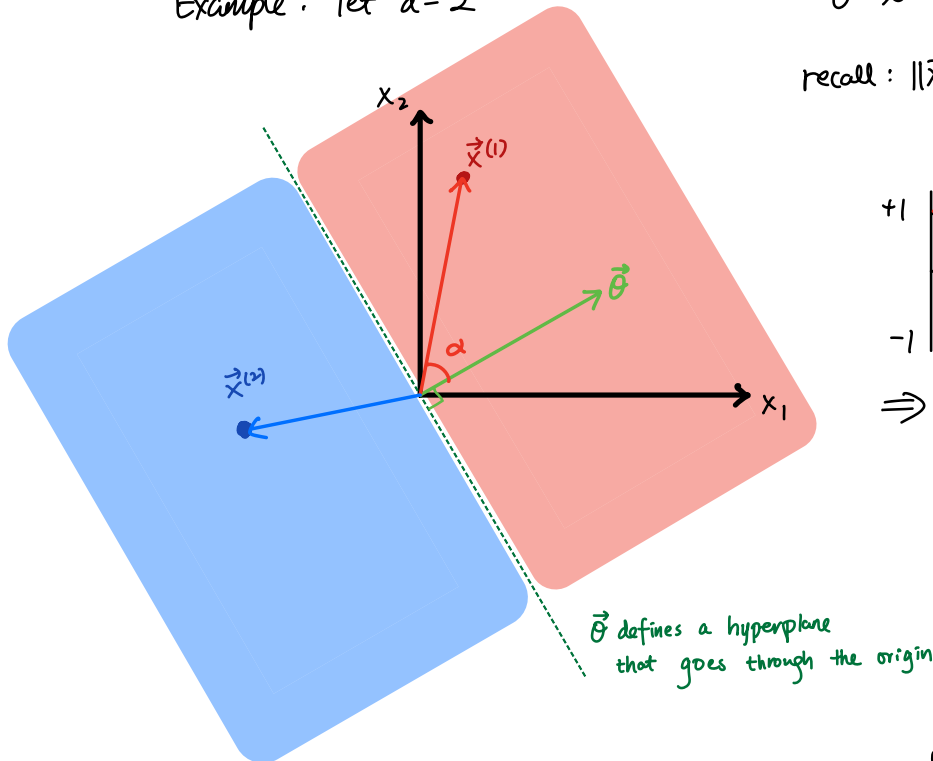
Thresholded linear mapping from feature vectors to labels

$$h(\vec{x}; \vec{\theta}) = \begin{cases} +1 & \text{if } \vec{\theta} \cdot \vec{x} > 0 \\ -1 & \text{if } \vec{\theta} \cdot \vec{x} < 0 \end{cases} \quad \text{where } \vec{\theta} \in \mathbb{R}^d$$
$$\vec{\theta} = [\theta_1, \theta_2, \dots, \theta_d]^T$$

$$\vec{\theta} \cdot \vec{x} = \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_d x_d \quad \leftarrow \text{linear combination of input features.}$$

Different $\vec{\theta}$'s produce (potentially) different labelings of the same $\vec{x}$.

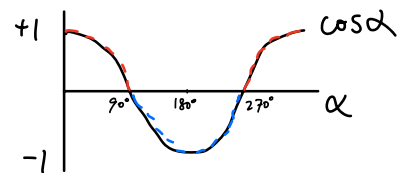
Example: let $d=2$



$$\vec{\theta} \cdot \vec{x} = \|\vec{\theta}\| \|\vec{x}\| \cos \alpha$$

$$\text{recall: } \|\vec{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2}$$

always ≥ 0



$\Rightarrow$ angle α determines the sign of $\vec{\theta} \cdot \vec{x}$

What if a point lies on the decision boundary?

$$\vec{\theta} \cdot \vec{x} = \|\vec{\theta}\| \|\vec{x}\| \cos 90^\circ = 0.$$

$\therefore$ the hyperplane can be defined as $\vec{\theta} \cdot \vec{x} = 0$.

in 2-dimensions, $\theta_1 x_1 + \theta_2 x_2 = 0$

$$\text{re-arranging, } x_2 = -\frac{\theta_1}{\theta_2} x_1$$

$\leftarrow$ "slope" of the decision boundary in 2D.

Does length matter?

Does direction matter?

Informally speaking,

$\vec{\theta}$ determines the "orientation" and "side" of the decision boundary

NOW: how do we select $\vec{\theta}$?

Intuition: find a $\vec{\theta}$ that works well on the training data $D = \{\vec{x}^{(i)}, y^{(i)}\}_{i=1}^N$

Wait, I thought you said overfitting? Why is it ok now?

→ We've restricted the class of possible classifiers to linear, reducing the chance of overfitting.

Minimize Training Error

Training error: fraction of training examples for which the classifier predicts the wrong labels.

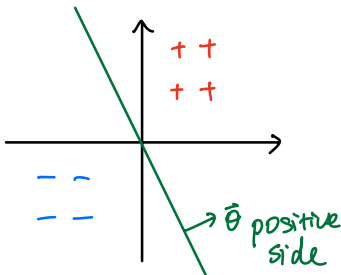
$$E_N(\vec{\theta}) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[y^{(i)} \neq h(\vec{x}^{(i)}, \vec{\theta})]$$

$\mathbb{1}[\cdot]$ returns 1 if logical expression $\cdot$ is true, 0 otherwise.

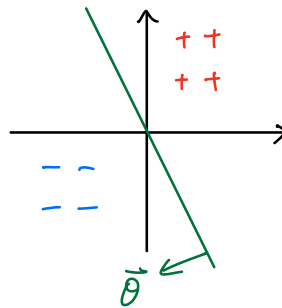
$$= \frac{1}{N} \sum_{i=1}^N \mathbb{1}[y^{(i)}(\vec{\theta} \cdot \vec{x}^{(i)}) \leq 0]$$

↗ when $y^{(i)}$ and $\vec{\theta} \cdot \vec{x}^{(i)}$ have opposite signs.

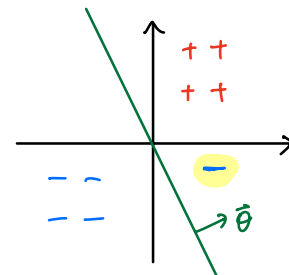
≤ by convention, points on the decision boundary are misclassified



training error = 0



training error = 1



training error = $\frac{1}{9}$

Next Lecture: Instead of trying all possible $\vec{\theta}$'s one by one, how can we efficiently "learn" the best $\vec{\theta}$?

→ In-Class Exercise #3.